Minsoo Song

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LIFELONG RESEARCH OBJECTIVE

My lifelong research objective is to make NLP reliable by evaluating evaluation itself. Benchmarks, leaderboards, and LLM judges are the instruments by which the field decides what language models can do, yet these instruments can themselves be flawed, biased, or fragile. I aim to audit and strengthen them, so that every claim about a model's capability rests on evidence that can be trusted.

RESEARCH INTERESTS

My research centers on evaluating evaluation as a path toward reliable NLP. I study whether the instruments we use to measure language models—benchmarks, leaderboards, and LLM judges—are themselves valid: identifying flawed benchmark items, auditing whether leaderboard rankings reflect genuine capability, and exposing biases and representational sensitivities in LLM-as-a-judge protocols. I am also interested in interpretability as a complementary lens, examining what models encode in their internal states to explain why evaluation outcomes arise.

EDUCATION

- **Soongsil University** Mar. 2027 - Aug. 2028 (Expected)
  M.S. in Software (advisor: Chanjun Park)
  Lab: https://sites.google.com/view/ssu-nlp/home
- **Soongsil University** Mar. 2021 - Feb. 2027 (Expected)
  B.S. in Software (advisor: Chanjun Park)
  GPA:
- **École 42 Seoul** Jan. 2022 - May 2022, Nov. 2023 - May 2024

WORK EXPERIENCE

- **Soongsil University** Sep. 2025 - Present
  Research Assistant
  Natural Language Processing Lab

PUBLICATIONS

Equal contribution: *, *Corresponding author*: †

Google Scholar: https://scholar.google.com/citations?user=G-G5t0AAAAAJ

International Conferences

1. **Structured but Silent: Probing Capability Requirements in LLM Hidden States**
Kyojun Choo, Minsoo Song, Yunju Kang, Chanjun Park†
Findings of the 2026 Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the International Joint Conference on Natural Language Processing (AACL-IJCNLP 2026)

2. **Auditing MCQA Benchmarks through Probability Landscapes**
Minsoo Song, Chanjun Park†
Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), Main Conference

3. **Beyond Consensus: Downward Bias and Role Asymmetry in Multi-Agent LLM Judges for Subjective Evaluation**
Minsoo Song, Chanwoo Kim, Sugyeong Eo†, Chanjun Park†
Findings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)

Domestic Conferences

1. **오답의 의미적 타당성 제거를 통한 LLM 벤치마크 결함 문항 식별**
(Flagging Flawed Benchmark Items by Removing Semantic Plausibility of Distractors)
Minsoo Song, Chanwoo Kim, Kyojun Choo, Eunseo Song, Giyoon Jang, Chailin Nam, Sihyeon Lee, Chanjun Park†
Proceedings of the 38th Annual Conference on Human and Language Technology (HCLT 2026)
Outstanding Paper Award

2. **다중 에이전트 토론을 활용한 언어모델의 한국어 추론 오류 진단**
(Diagnosing Korean Reasoning Errors in Language Models Using Multi-Agent Debate)
Jiwoo Kum, Minsoo Song, Chanwoo Kim, Sihyeon Lee, Doyun Kim, Chanjun Park†
Proceedings of the 38th Annual Conference on Human and Language Technology (HCLT 2026)

3. **행동적 타당성 감사를 통한 리더보드 강건성 분석 연구**
(A Study on Leaderboard Robustness Analysis through Behavioral Validity Auditing)
Kyojun Choo, Minsoo Song, Seonghyeon Cho, Giyoon Jang, Chailin Nam, Chanjun Park†
Proceedings of the 38th Annual Conference on Human and Language Technology (HCLT 2026)

4. **대규모 언어 모델의 도구 호출에서의 이름 의존성 분석**
(Name Dependence in Large Language Model Tool Calling)
Chailin Nam, Seonghyeon Cho, Sihyeon Lee, Jiwoo Kum, Minsoo Song, Eunseo Song, Chanjun Park†
Proceedings of the 38th Annual Conference on Human and Language Technology (HCLT 2026)

5. **한국어 LLM Judge는 의미를 보존하는가? 척도와 응답 순서에 대한 표현 불변성 분석**
(Do LLM Judges Preserve Meaning in Korean Evaluation? An Analysis of Representational Invariance in Scale Direction and Response Order)
Chanwoo Kim, Doyun Kim, Minsoo Song, Sihyeon Lee, Jihun Song, Chanjun Park†
Proceedings of the 38th Annual Conference on Human and Language Technology (HCLT 2026)

AWARDS

- **Outstanding Paper Award**, The 38th Annual Conference on Human and Language Technology (HCLT 2026)
  오답의 의미적 타당성 제거를 통한 LLM 벤치마크 결함 문항 식별

RESEARCH PROJECTS

Soongsil University

- 2026년 인공지능 말평 운영 및 연계 행사 개최 (2026, 참여연구자(PM) (용역과제), National Institute of the Korean Language)
- 2026년 인공지능의 한국어 능력 표준 평가 과제 구축 및 활용 (2026, 참여연구자 (용역과제), National Institute of the Korean Language)
- 우수연구 신진연구 (유형B) (2026-2029, 참여연구자 (3책5공), NRF)
- Foundation Model 운용과정에서 민감정보 추론 방지 및 리스크 평가 기술 개발 (2026-2028, 참여연구자 (3책5공), Korea Internet & Security Agency (한국인터넷진흥원))
- 토론과 소통으로 지식을 연결하는 AI 에이전트 기술 개발 (2026-2028, 참여연구자 (3책5공), Korea Creative Content Agency (한국콘텐츠진흥원))
- AI 보안 특화 'AI 에이전트' 개발 선행 연구 (2026, 참여연구자 (용역과제), 한국정보보호학회 산하 AI보안연구회)
- 국가 · 공공기관 가이드라인 준수를 위한 AI 시스템 보안 위협 자동화 기술점검 도구 개발 (2026, 참여연구자, KIST)
- AI 연구용 컴퓨팅 지원 프로젝트 (파라미터 수준의 정보 제어 원천기술 연구 및 파라미터 스튜디오 플랫폼 개발) (2026, 참여연구자, KETI)

RESEARCH GRANTS/FELLOWSHIPS

Research Grants

- VESSL AI GPU Grant Program (2026)

PATENTS

International Patents

Domestic Patents

- **인공지능 모델 평가용 벤치마크를 분석하는 전자 장치 및 이의 동작 방법**
  박찬준, 송민수
  Korean Patent Application, filed on August 4, 2026 (Application No. 10-2026-0144925).
- **인공지능 평가 프로토콜을 분석하는 전자 장치 및 이의 동작 방법**
  박찬준, 송민수
  Korean Patent Application, filed on August 4, 2026 (Application No. 10-2026-0144924).

EXTERNAL ACTIVITIES

Teaching Assistant

- SaRangAXConnect, AI Transformation (AX) Program (September 2026)